THREE PARADIGMS OF PHYSICAL WORLD MODELING

How should a Physical AI agent represent the environment, predict future dynamics, and survive sensory occlusion?

1. GENERATIVE PIXELS

Pixel-Space Diffusion / Autoregression

Observation Space:
Raw Pixels $\hat{I}_{t+1} \in \mathbb{R}^{H \times W \times 3}$
Forward Prediction:
Denoising U-Net / Video DiT
$\hat{I}_{t+1} = \mathcal{D}_\theta(\epsilon, I_t, a_t)$

Embodied Failure Modes:

- **Compute Tax:** > 30 TFLOPs/frame
- **Latency Wall:** 300–1500 ms $\gg \tau_{\text{world}}$
- **Visual Drift:** Non-rigid morphing
- **Non-Metric:** No $SE(3)$ frame units

UNVIABLE FOR REAL-TIME

2. LATENT DYNAMICS

Non-Generative JEPAs & RSSMs

Representation Space:
Latent Features $z_t \in \mathbb{R}^D$ $(D \ll HW)$
Forward Prediction:
Latent Transition $z_{t+1} = g_\phi(z_t, a_t)$
Recurrent h_t + Stochastic z_t

Embodied Advantages:

- **Sub-Millisecond:** $t_{\text{rollout}} < 2.5$ ms
- **Noise Invariance:** Discards clutter
- **Multi-Step MPC:** Rollout $H = 32$
- **Direct Planning:** Optimizes in $\mathcal{Z}$

OPTIMAL FOR INTENT PLANNING

3. METRIC $SE(3)$ TREES

Rigid Lie Algebra Frame Graphs

State Space:
Homogeneous $T_B^A \in SE(3)$, $\boldsymbol{\xi} \in \mathfrak{se}(3)$
Forward Prediction:
Time-Indexed Graph + EKF
$T(t + \Delta t) = T(t) \exp(\xi^\wedge \Delta t)$

Deterministic Guarantees:

- **Microsecond Query:** $< 10\,\mu$s lookup
- **Metric Persistence:** Occlusion tracking
- **Uncertainty Bounds:** $\sigma_{\text{pos}}(t)$ growth
- **Safety Invariant:** Collision bounding

MANDATORY FOR SAFETY

THE HYBRID STATE ARCHITECTURE OF PHYSICAL AI

The agent combines **Metric $SE(3)$ Frame Trees** (deterministic 3D spatial persistence on MCU/MPU) with **Latent World Models** (fast semantic feature rollouts on NPU), completely bypassing slow, hallucination-prone pixel diffusion.